

Onboarding Guide for Engineers

Synthetic Company Edition • Fictional training and demonstration document

Welcome, [Engineer's Name]! We're excited to have you join **Northstar Labs**. You are now part of a team of engineers, designers, QA specialists, and infrastructure professionals building reliable software products for our customers.

This guide is designed to help you understand how the engineering organization works, what you should do during your first week, where to find important resources, and who to contact when you need help.

Important Note

This is a fully synthetic example. Northstar Labs, the people, projects, handles, repositories, and links in this document are fictional and are included only for demonstrations, testing, and training.

1. Welcome Message

Welcome to the engineering team! Your first few days will focus on getting comfortable with the people, tools, development workflow, and product areas that you will work with.

You are not expected to understand everything immediately. Ask questions, document what you learn, and use your mentor and project team as your primary sources of guidance.

Our Engineering Principles

- **Build with purpose:** Understand the user and business problem before writing code.
- **Keep it simple:** Prefer clear, maintainable solutions over unnecessary complexity.
- **Own your work:** Take responsibility for quality, documentation, testing, and follow-up.
- **Collaborate openly:** Ask for reviews early and communicate blockers quickly.
- **Learn continuously:** Treat feedback and mistakes as opportunities to improve.

2. Team Structure

Our engineering organization is divided into several functional groups. Teams work closely together through shared planning meetings, code reviews, design reviews, and incident-response processes.

Role	Name	Handle	Primary Responsibility
Engineering Manager	Arjun Menon	@arjun-memon	Engineering strategy, delivery, people management
Backend Lead	Priya Nair	@priya-nair	Backend architecture and API standards
Frontend Lead	Rohan Kapoor	@rohan-kapoor	Web architecture and UI engineering
QA Lead	Vikram Das	@vikram-das	Test strategy and release quality
DevOps Lead	Neha Verma	@neha-verma	Cloud, CI/CD, infrastructure
UX/UI Lead	Simran Kaur	@simran-kaur	Product design and user experience
Data Lead	Kavita Shah	@kavita-shah	Data pipelines and analytics

When asking a question, use the appropriate team channel first. If the issue is urgent or blocks delivery, contact your project lead or mentor directly.

3. Project Overview

New engineers are typically assigned to one primary product area and may contribute to shared engineering initiatives. Below are the current synthetic projects.

Project	Purpose	Primary Area	Lead
Atlas	Employee onboarding platform	Backend	Priya Nair
Pulse	Product analytics dashboard	Frontend/Data	Rohan Kapoor
Forge	Developer environment automation	DevOps	Neha Verma
Shield	Automated QA and regression testing	Quality	Vikram Das
Nova	Customer notification service	Backend	Priya Nair
Orbit	Cloud infrastructure automation	Infrastructure	Neha Verma
Beacon	Application monitoring and observability	Platform	Arjun Menon
Lumen	Business intelligence reporting suite	Data	Kavita Shah

Your First Project

Your project lead will schedule a project deep dive covering the product goal, architecture, current sprint, repository structure, development environment, and known technical risks.

Before the deep dive, review the project's README, architecture documentation, issue tracker, and most recent release notes.

4. Key Resources

- **Code Repository:** <https://example.com/northstar/engineering>
- **Engineering Documentation:** <https://example.com/northstar/docs>
- **Issue Tracker:** <https://example.com/northstar/issues>
- **Design Workspace:** <https://example.com/northstar/design>
- **CI/CD Dashboard:** <https://example.com/northstar/ci>
- **Monitoring Dashboard:** <https://example.com/northstar/monitoring>
- **HR Portal:** <https://example.com/northstar/hr>
- **Learning Portal:** <https://example.com/northstar/learning>

All URLs above are fictional example addresses. Replace them with the actual internal links when adapting this guide for a real organization.

5. Engineering Technology Stack

Our synthetic engineering environment uses a modern full-stack toolset. Your exact stack depends on your project.

- **Frontend:** React, TypeScript, HTML, CSS
- **Backend:** Python, FastAPI, Node.js
- **Databases:** PostgreSQL and Redis
- **Data:** Python, SQL, scheduled data pipelines
- **Testing:** Pytest, Jest, Playwright
- **Version Control:** Git and GitHub-style pull requests
- **CI/CD:** Automated build, test, and deployment pipelines
- **Infrastructure:** Containers and cloud-based services

Recommended Development Setup

- Install Git and configure your work identity.
- Install the approved code editor or IDE.
- Install the project's required runtime and package manager.
- Configure environment variables using the team's setup instructions.
- Clone the assigned repository.
- Run the project's test suite locally.
- Run the application locally and verify the health endpoint.
- Confirm that you can create a branch and open a draft pull request.

6. First Week Agenda

Day	Focus	Activities
Day 1	Orientation and Setup	HR orientation, workstation setup, accounts, mentor introduction.
Day 2	Technology and Product	Tech-stack walkthrough, product overview, development workflow.
Day 3	Project Deep Dive	Architecture review, repository walkthrough, first project assignment.
Day 4	Engineering Workflow	Stand-up, sprint planning, code review, testing and deployment process.
Day 5	Review and Next Steps	Progress review, questions, feedback, and plan for the second week.

First-Week Checklist

- Complete HR paperwork.
- Set up your development environment.
- Join the engineering stand-up.
- Meet your mentor.
- Attend the technology-stack presentation.
- Read the engineering contribution guide.
- Read your project's README.
- Run the project locally.
- Complete security and access training.
- Open your first draft pull request.
- Attend the end-of-week review.

7. Engineering Workflow

Daily Stand-Up

Stand-up is a short synchronization meeting. Share what you completed, what you are working on next, and anything blocking progress. Avoid turning stand-up into a long technical discussion; move detailed conversations to a follow-up meeting or team channel.

Branches and Pull Requests

- Create a short-lived branch for each change.
- Use a descriptive branch name such as `feature/onboarding-dashboard`.
- Keep commits focused and easy to understand.
- Run tests before requesting review.
- Open a pull request with a clear summary and testing notes.
- Address review comments respectfully and document important decisions.

Example Git Workflow

```
git checkout -b feature/example-change
git add .
git commit -m "Add example change"
git push -u origin feature/example-change
```

8. Quality Assurance and Testing

Quality is shared across the engineering organization. Developers are responsible for writing maintainable code and appropriate tests, while QA helps define broader validation strategies.

- **Unit tests:** Verify individual functions or components.
- **Integration tests:** Verify interactions between multiple components or services.
- **End-to-end tests:** Verify important user workflows.
- **Regression testing:** Confirm that existing behavior still works after changes.
- **Code review:** Validate correctness, maintainability, security, and consistency.

Before Opening a Pull Request

- Run formatter and linter.
- Run relevant unit tests.
- Run integration tests when applicable.
- Review your own diff.
- Update documentation if behavior changed.
- Add screenshots for meaningful UI changes.
- Mention known limitations or follow-up work.

9. Security and Access

Security is everyone's responsibility. Never place credentials, API keys, passwords, tokens, or private customer data in source code or public issue comments.

- Use approved secret-management mechanisms.
- Lock your workstation when away.
- Do not share credentials through chat.
- Report suspicious access or security issues immediately.
- Use only the minimum permissions required for your work.
- Do not copy production customer information into development environments.

Security Checklist

- Complete security awareness training.
- Enable multi-factor authentication where required.
- Verify access to only the systems needed for your role.
- Review the secure coding guidelines.
- Know how to report a security incident.

10. Communication Guidelines

Good engineering communication is concise, searchable, and respectful.

- **Urgent:** Contact the relevant lead directly and post a short summary in the appropriate channel.
- **Blocker:** Mention it during stand-up and notify your mentor or project lead.
- **Technical question:** Search documentation first, then ask in the engineering channel.
- **Design question:** Include the relevant UX/UI team member.
- **Production issue:** Follow the incident-response procedure rather than attempting an undocumented fix.

Synthetic Team Channels

- #engineering-general — general engineering discussions
- #engineering-help — technical questions and troubleshooting
- #project-atlas — Atlas project discussions
- #project-pulse — Pulse project discussions
- #devops-support — infrastructure and deployment questions
- #qa-testing — testing and quality discussions

11. Mentor Details

Your mentor is your primary point of contact during the onboarding period. They can help you understand engineering processes, team culture, project context, and where to find information.

Mentor	Role	Experience	Focus
Priya Nair	Backend Lead	6 years	APIs, services, architecture
Rohan Kapoor	Frontend Lead	4 years	React, TypeScript, UI architecture
Vikram Das	QA Lead	7 years	Testing and release quality
Neha Verma	DevOps Lead	5 years	CI/CD, cloud and infrastructure
Simran Kaur	UX/UI Lead	3 years	Design systems and UX

Mentor	Role	Experience	Focus
Kavita Shah	Data Lead	6 years	Analytics and data engineering

Suggested Mentor Meetings

- Day 1: 30-minute introduction and onboarding expectations.
- Day 3: 45-minute project and architecture discussion.
- Day 5: 30-minute first-week review.
- Week 2 onward: Weekly 30-minute check-in.

12. Expectations During Your First 30 Days

Week 1

- Understand the organization and project
- Set up your environment
- Meet key teammates

Weeks 2–3

- Complete a small production-quality task
- Participate in code reviews
- Learn the team's testing and deployment workflow

Week 4

- Own a small feature or improvement
- Present what you learned
- Agree on development goals with your manager

13. Frequently Asked Questions

Who should I ask when I am stuck?

Start with your documentation and project resources. If you still need help, ask your mentor or the appropriate engineering channel. If you are blocked on a delivery-critical issue, contact your project lead.

How much should I ask during onboarding?

Ask early rather than spending hours stuck on an issue that someone could clarify in a few minutes. Good questions include the context, what you tried, the result, and what you think should happen.

What if I make a mistake?

Mistakes happen. Communicate them quickly, help resolve the impact, and document what you learned. The goal is to improve the system and the process rather than hide problems.

Where do I find project-specific instructions?

Start with the project README and documentation. Your project lead or mentor can point you to additional resources.

14. First-Month Completion Checklist

- Completed all required HR and security onboarding.
- Configured the local development environment.
- Joined the engineering stand-up and relevant team channels.
- Met your mentor, manager, and project lead.
- Read the engineering contribution guidelines.
- Read the assigned project's documentation.
- Successfully ran the application locally.
- Submitted at least one pull request.
- Participated in a code review.
- Participated in testing or QA validation.
- Understood the deployment process.
- Completed a small engineering task.
- Attended the first-month review.

15. Welcome to the Team

We are glad to have you here. Take ownership, stay curious, communicate clearly, and help the people around you. Your first month is about learning how the team works; the months after that are about using what you learned to make meaningful contributions.

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